

Homework 10

Please read Example 1.12, Example 1.23, 1.28 in Chapter 1 of Durrett. These examples all go together and describe the geometric distribution. Since a geometrically distributed random variable has infinitely many possible values (1, 2, 3, ...), calculations with such a random variable may require Calculus 2 techniques involving infinite series. The key infinite series for this type of random variable is the “geometric series,” which is the Taylor series expansion for $\frac{1}{1-x}$ as a function of x : $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$. The series converges for x between -1 and 1 . Durrett uses this series and related infinite series for $x=p$, the probability of success in each trial.

Please reread Section 2.3 through the end of Theorem 2.3. To calculate with the Poisson distribution, Durrett uses the Taylor series expansion for e^x as a function of x :

$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots$. This series converges for any x . Durrett uses this series and related infinite series for $x=\lambda$, the parameter for a Poisson distribution.

Please also read Section 2.4 through Example 2.41. Durrett provides many examples like the first question on Quiz 4. He uses the name “urn problems” for questions like this.

1. I hope that in Homework 9 question 6, you proposed $N \sim \text{Poisson}(1)$ as a model for the number of misprints on a random page of a book from the publisher. We have $P(N \geq 5) = 1 - P(N \leq 4)$. Using the formula for the Poisson(1) distribution, we have the following:

$$P(N \leq 4) = P(N=0) + P(N=1) + P(N=2) + P(N=3) + P(N=4) = e^{-1} + e^{-1} + \frac{1}{2}e^{-1} + \frac{1}{6}e^{-1} + \frac{1}{24}e^{-1} = \frac{65}{24}e^{-1}.$$

Therefore, $P(N \geq 5) = 1 - \frac{65}{24}e^{-1}$.

- a. Imagine a 300-page book. Assume that errors on different pages are independent. What (binomial) distribution would you use for the random variable counting the number of pages with at least 5 misprints?

Let X be the random variable counting the number of pages with at least 5 misprints. The probability of at least 5 misprints on any page is $1 - \frac{65}{24}e^{-1}$. Assuming that errors on different pages are independent, we have $X \sim \text{binomial}\left(300, 1 - \frac{65}{24}e^{-1}\right)$.

- b. Use the Poisson approximation to estimate the probability that in the book there is at least one page with at least 5 misprints.

The Poisson approximation tells us that $\text{binomial}\left(300, 1 - \frac{65}{24}e^{-1}\right)$ probabilities are approximately $\text{Poisson}\left(300 \cdot \left(1 - \frac{65}{24}e^{-1}\right)\right)$ probabilities. We have $300 \cdot \left(1 - \frac{65}{24}e^{-1}\right) \approx 1.097954$, and so $P(X \geq 1) = 1 - P(X = 0) \approx 1 - e^{-1.098} = 0.6664$.

2. You started thinking about the following situation in class. Imagine an unreliable clothes dryer that does one of two things when you put money in:
- It takes 20 minutes to dry your clothes. This event occurs with probability 0.6.
 - It buzzes for 4 minutes and does nothing to your clothes. This even occurs with probability 0.4.

You plan to keep putting money into the machine until it dries your clothes. Assume that successive trials with the dryer are independent.

- a. Let N be the random variable that counts how many times you have to run the dryer until it dries your clothes. What distribution does N have? Use the “~” notation and the name of the distribution.

The story of the clothes washer matches the geometric distribution. We have repeated independent trials (trying to dry our clothes) with probability of success 0.6. We thus have $N \sim \text{geometric}(0.6)$.

- b. The random variable $N - 1$ counts the number of times that the dryer fails before its first successful run. Therefore, $4(N - 1)$ is the amount of time the dryer spends on unsuccessful runs. Calculate the expected time $E[4(N - 1)]$. Use the shortcuts you read about in Chapter 1: Theorem 1.1 (right after Example 1.19) and the two properties numbered as Equation 1.10 (right after Example 1.25).

Using the properties of expected value from Durrett, we have the following:

$$E[4(N - 1)] = 4E[N - 1] = 4(E[N] - 1) = 4 \cdot \left(\left(\frac{1}{0.6}\right) - 1\right) \text{ using the formula } E[N] = \frac{1}{p} \text{ for a geometric } (p) \text{ random variable.} \approx 2.67 \text{ minutes.}$$

- c. What is the expected time to dry your clothes?

The expected wait time before the dryer works is 2.67 minutes. It takes 20 minutes to dry the clothes. Therefore, the expected time to dry the laundry is $2.67 + 20$ minutes or 22.67 minutes.

Please work the following additional exercises from **Chapter 2**.

The product of a defect in a foot of magnetic tape—do you know what Durrett is talking about? —is 0.002. If we think of a 1500 foot roll of tape as 1500 independent one-foot segments, we can model the total number of defects using the binomial(1500, 0.002) distribution. This distribution is approximately the Poisson(λ) distribution for $\lambda = 1500(0.002) = 3$. Therefore $P(\text{no defects}) \approx e^{-3} \approx 0.04979$.

52 (red and black cards)

We solve the problem using the “urn problems” approach. To say that the two cards we draw are different colors is to say that we draw one red and one black. The number of ways to draw one red and one black is $C_{2,1} C_{2,1} = (2)(2) = 4$. The number of possible combinations of two cards drawn from the set of 4 is $C_{4,2} = 6$. Therefore, the probability of two different colors is $4/6$ or $2/3$.

53 (textbook answer: 10/21. Explain how to get this probability using the “urn problems” approach from Quiz 4 and the reading.)

We assume that the couples are man/woman pairs. (Durrett probably came up with this question in the 1990s, when almost everyone would make that assumption without thinking.) We thus have an “urn” with five men and five women. We can count the number of groups of four people with two men and two women using the multiplication rule: First choose 2 of the 5 men and then 2 of the 5 women. There are $C_{5,2} C_{5,2} = (10)(10) = 100$ possible choices. The number of 4-person groups that we can choose out of the 10 people is $C_{10,4} = (10)(9)(8)(7)/24 = 210$. Assuming that all the 4-people groups are equally likely, the probability of choosing 2 men and 2 women is $100/210 = 10/21$.

54 (this kind of question should be sounding familiar)

We pick 5 cards at random from a deck of 52. (That’s what we usually call a “poker hand.”) We think of the deck as an “urn” with 52 cards in it. There are 13 spades in the deck/urn. We use the multiplication rule to count the number of 5-card hands with exactly two spades: We first pick 2 spades cards and the 3 non-spades cards. There are $C_{13,2} C_{39,3}$ possible choices. The number of 5-card hands is $C_{52,5}$. Assuming all the 5-card hands are equally likely, the probability of a hand with exactly 2 spades is $C_{13,2} C_{39,3} / C_{52,5}$. That works out to be about 0.2743.